

CellMap Process Book

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1. Overview

1.1. Motivation

The rapid advancement of medical imaging technologies has significantly transformed the landscape of healthcare diagnostics and research. Histopathology, the study of microscopic changes in tissues caused by disease, remains a cornerstone in diagnosing numerous medical conditions, including cancers. Traditionally, histopathological analysis has relied heavily on manual inspection by trained pathologists who examine tissue samples under a microscope.

This conventional approach, while effective, poses several challenges. It is inherently time-consuming, labor-intensive, and subject to inter-observer variability, which can lead to diagnosis inconsistencies. Moreover, the increasing volume of medical data due to improved imaging techniques demands more efficient methods for analysis and interpretation.

Digital pathology has emerged as a promising solution by digitizing tissue samples into high-resolution images that can be analyzed computationally. This transition has opened up new avenues for integrating computational tools, such as machine learning and data visualization, into the diagnostic process. Machine learning algorithms, particularly deep learning models, have demonstrated remarkable success in image recognition tasks and are increasingly being applied to medical image analysis. These models can assist in identifying patterns and features that may not be easily discernible to the human eye, thereby augmenting the diagnostic capabilities of pathologists.

Despite these advancements, a critical hurdle remains the practical interpretation and utilization of the complex data generated by computational models. Raw outputs from machine learning algorithms are often immediately interpretable only by clinicians with a data science background. Therefore, there is a pressing need for tools to bridge this gap by translating complex computational outputs into intuitive and interactive visualizations that medical professionals can easily understand and utilize.

The specific focus on breast cancer histopathology is driven by the significant impact of breast cancer worldwide. Breast cancer is one of the most common cancers among women globally, and early and accurate diagnosis is critical for effective treatment. Enhancing the tools available for histopathological analysis of breast tissue can, therefore, substantially impact patient outcomes. This project aims to apply technical skills to real-world problems with meaningful implications for healthcare.

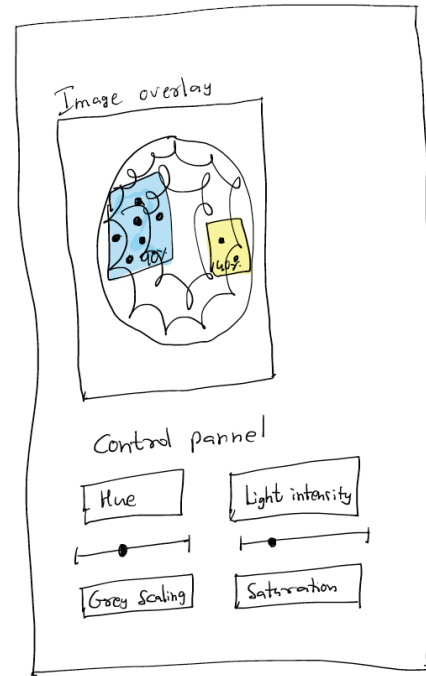


Figure 1. Project Plan Draft

1.2. Project Plan

By harnessing the power of data visualization, the primary objective of this project is to develop an interactive visualization tool that enhances the analysis of histopathological images by integrating machine learning predictions. Specifically, by displaying the prediction results of a model over images reassembled from individual image tiles, the tool aims to produce intuition to enable pathologists and researchers to better understand and analyze the model.

2. Related Work

2.1. Breast Cancer Detection with CNN

Deep learning techniques, particularly convolutional neural networks (CNNs), have proven to be highly effective in medical image analysis, including breast cancer detection. The use of CNNs enables automatic feature extraction from medical images, bypassing the need for manual feature engineering, and achieving state-of-the-art performance in classification and diagnosis tasks.

Studies leveraging publicly available datasets, such as the Wisconsin Breast Cancer Dataset (WBCD) or mammogram datasets, have shown that CNNs can accurately differentiate between malignant and benign cases [2]. This approach highlights the challenges and opportunities asso-

ciated with applying CNNs in real-world medical diagnostics, thus understanding the performance of the models as detailed as possible is crucial.

3. Data

3.1. Dataset Description

The Breast Cancer Histopathological Image Dataset, available on Kaggle [1], comprises 277,524 histopathological images of breast tissue collected at 40x magnification. The images are categorized into two classes:

- **Invasive Ductal Carcinoma (IDC) Positive:** Images containing malignant cells.
- **IDC Negative:** Images without malignant cells.

Each image is a 50x50 pixel patch extracted from whole-slide images (WSIs), with corresponding labels indicating the presence or absence of IDC. The dataset provides a rich source of information for training and validating machine learning models aimed at detecting breast cancer.

3.2. Model Retrieval

The application is designed to visualize the results of machine learning models, such as convolutional neural networks (CNNs). For demonstration purposes, we have adapted and modified a publicly available model at [this link\[2\]](#). The model is retrained on the dataset and subsequently saved in an .h5 format for future use. Since the primary focus of this project is on data visualization rather than model performance, the training process has been deliberately simplified, and suboptimal prediction performance is anticipated.

4. Preprocessing & Exploratory Data Analysis

4.1. Image Display

To present the initial visualization, the images need to be reassembled from individual tiles. (Figure.2)

Each tile file is named using the format: case_id_idx5_x-coordinate_y-coordinate, which encodes its association with a specific case and its position within the overall image. By following this naming convention, the tiles can be systematically collected and reconstructed into their original images for visualization purposes. Additionally, since tiles with different labels are stored in separate folders, annotating positive tiles becomes straightforward, further simplifying the visualization and analysis process. (Figure.3)

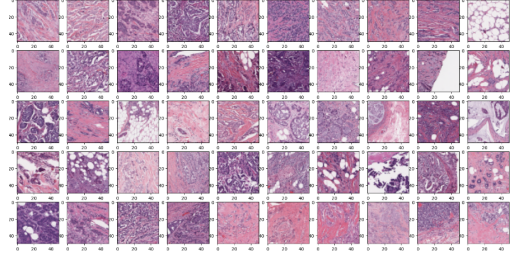


Figure 2. Image Tiles

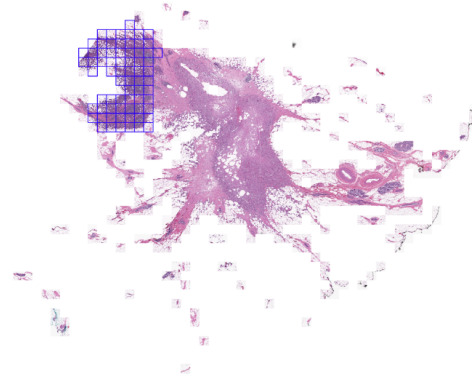


Figure 3. Annotated Full Image

4.2. Produce the annotation

By loading the previously generated .h5 model file, we are prepared to simulate the user's model training process. To narrow the scope for demonstration purposes, we focus on case 10253. Predictions are generated for each tile associated with this case, and the results are saved in a .json format to facilitate future use and integration with the visualization tool.

5. Questions

We aim to assist users in understanding the model's performance by addressing the following key questions:

5.1. Overall Model Performance

We evaluate the model's accuracy to provide an overall assessment of how well the model classifies tiles across the dataset, highlighting its effectiveness and areas for potential improvement.

5.2. Spatial Performance Patterns

To analyze spatial performance, we investigate whether the model exhibits patterns or biases in specific areas of the image. Prediction outcomes are visualized as overlays on

the image, allowing users to identify regions with consistent errors or areas where the model demonstrates high confidence in its predictions.

5.3. Confusion Matrix Analysis

The confusion matrix and related metrics are examined to uncover systematic trends or biases in the model's predictions. This includes exploring whether the model is prone to certain types of errors in specific regions or under particular conditions.

By addressing these aspects, users can better assess the model's reliability and interpretability, ensuring its outputs are meaningful and actionable in practical applications.

6. Design Evolution

6.1. Deprecated Features

During the development process, several initial features were deprecated as they were deemed non-critical to the primary objective of this data visualization project. Additionally, these features consumed excessive development time without providing proportional value. The deprecated features include:

1. **Multi-case Upload Support:** The ability for users to upload multiple cases simultaneously was removed to streamline the application and focus on visualizing individual cases effectively.
2. **Embedded Model Training and Testing:** Integrating the model training and testing process within the app was excluded, as it diverted resources from the primary objective of creating an intuitive and efficient visualization tool.
3. **Model Upload Option:** Allowing users to upload custom models was removed to simplify the app's workflow and avoid potential compatibility and user experience issues.

6.2. Missing Features

While the project achieved its primary objectives, there are certain features that were not finalized due to time constraints. These missing features, which hold significant potential for enhancing user experience and functionality, include:

1. **Control Panel for Case Management:** A control panel that grants users greater control over selecting

and displaying specific cases. This would improve interactivity and allow users to navigate between cases seamlessly.

2. **Tile Selection with Dynamic Chart Updates:** The ability for users to select specific tiles and dynamically update supportive charts based on their selection. This feature would provide deeper insights and interactivity, enabling users to explore relationships and patterns in the data more effectively.

These features are envisioned as part of future work to further refine and extend the functionality of the application.

6.3. Evolved Features

Please notice that as the initial proposal did not include specific detailed design commitments, no comparisons nor discussion of design evolution can be presented. For a comprehensive overview of the features included in the final version, please refer to the **Implementation** section.

7. Implementation

The implementation contains two files, a `index.html` as the home page and a `script.js` called by it. The visualization mainly contains three parts: a main canvas, a sub-canvas, and a supportive popup panel.

7.1. Main Canvas

The main canvas serves as the core component of the visualization. Upon uploading the data, the main canvas displays the complete image of the selected case and provides zoom-and-pan functionality, which also controls the sub-canvas for synchronized navigation. Also, the users can decide whether to turn off the accuracy mask or not.

In the main canvas, block-like tiles are used as visual markers, encoding multiple attributes:

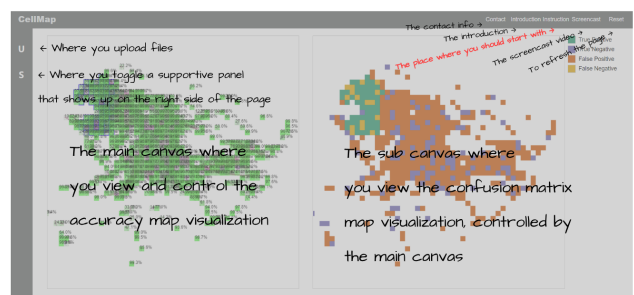


Figure 4. Overall Implementation

- **Location Channel:** Represents the physical location of each tile within the case sample.
- **Color Channel (of the mask):** The predicted class of each tile is represented using a Blue-Yellow-Red colormap, where **Blue** indicates a 100% probability of being predicted as positive, **Red** indicates a 100% probability of being predicted as negative, and **Yellow** corresponds to a balanced probability of 50%-50%. The intermediate colors between these points are determined using a linear gradient to ensure smooth and accurate visual transitions. This colormap has been proven to be color-blind friendly, To enhance the visibility of the underlying image, an initial opacity of 50% is applied to the colormap overlay. Users are provided with the ability to adjust the opacity dynamically through a drag bar.
- **Border Color:** Indicates the ground truth for each tile, with positive tiles highlighted by a blue border.

This visualization is designed to provide users with an "as-quick-as-possible" intuitive and comprehensive view of any regional patterns in the model's performance, enabling effective analysis and insight into its predictions.

7.2. Sub Canvas

The sub-canvas shows a confusion matrix map, mirroring the scope of the main canvas, with its zoom-and-pan functionality synchronized to the main canvas to ensure consistency in navigation and user experience.

In the sub-canvas, blocks are used as visual markers, where each block represents a tile. These blocks encode two primary attributes:

1. **Color:** Indicates the confusion matrix status of the corresponding tile, as defined in the accompanying legend. The color scheme has been carefully selected to pass color-blind accessibility tests.
2. **Position:** Reflects the spatial location of the tile within the image, maintaining alignment with the original structure.

This visualization is designed to enable users to identify and analyze regional performance patterns of the model in the current case, providing deeper insights into potential trends or anomalies in the predictions.

7.3. Supportive panel

Upon clicking the "Switch" sidebar, a supportive panel pops up from the right side of the page (Figure.5), overlay-



Figure 5. Supportive Panel Displayed

ing the area previously occupied by the sub-canvas. This panel provides three distinct information windows:

1. **Case Statistics Window:** Displays direct statistics of the uploaded case.
2. **Confusion Matrix Window:** Presents the confusion matrix summarizing the model's performance for the entire case.
3. **Bar Chart Window:** Visualizes derived metrics from the confusion matrix, offering additional insights into the model's predictions.

The panel operates in a mutually exclusive manner with the sub-canvas, as both aim to display the same aspect of the data but from different perspectives. While the sub-canvas focuses on spatial visualization, the panel provides a high-level overview through statistical and analytical representations. This complementary design allows users to gain a comprehensive understanding of the case.

8. Evaluation

8.1. Learnings from the Data

Through the visualizations, we gained a deeper understanding of the model's performance on histopathological image tiles. Key insights included the ability to detect regions where the model struggled to make confident predictions and identify patterns in misclassifications. The confusion matrix revealed systematic trends, such as higher false positive rates in specific areas, which may correspond to image artifacts or challenging regions in the dataset.

8.2. Answers to Questions

The visualizations effectively answered the research questions by:

- Demonstrating the overall accuracy of the model through clear statistics and performance metrics.

- Highlighting spatial patterns in model performance via the main and sub-canvas visualizations, allowing for regional analysis of predictions.
- Providing detailed insights into the confusion matrix and derived metrics, which clarified trends in false positives and false negatives across the dataset.

8.3. Assessment of Visualization

The visualizations proved effective in making the complex output of machine learning models accessible and interpretable. Strengths included:

- A user-friendly interface that allowed seamless navigation and exploration of the data.
- Comprehensive visual encoding, such as tile opacity for prediction confidence and border colors for ground truth comparison.
- The complementary use of the supportive panel and sub-canvas to provide both high-level and detailed views of the data.

However, certain areas could be improved:

- The responsiveness of the zoom-and-pan functionality could be further optimized for smoother navigation.
- More interactive elements, such as tooltips for individual tiles, could enhance the depth of insights available to users.

8.4. Future Improvements

Given more time, as mentioned in the deprecated and missing features sections, several enhancements could be implemented:

- Adding a feature for users to dynamically filter tiles based on prediction confidence or specific confusion matrix categories.
- Enhancing interactivity by allowing users to annotate regions of interest directly within the visualization.
- Expanding the tool to support multi-case analysis, enabling users to compare patterns across different cases more efficiently.

References

- [1] Paul Mooney. Breast histopathology images dataset, 2018. Accessed: 2024-12-02. [3](#)
- [2] Zead Omar. Breast cancer detection with cnn, 2022. Accessed: 2024-11-27. [2](#), [3](#)